Aadit Kshirsagar

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EDUCATION

University of Southern California — Aug 2025 - May 2027
Master of Science in Computer Science – Artificial Intelligence

Savitribai Phule Pune University — Aug 2020 - Jun 2024
Bachelor of Engineering in Computer Science — *GPA:8.89/10*

TECHNICAL SKILLS

- **Programming**: Python, Java, SQL, Git
- **Databases**: PostgreSQL, MongoDB, Chroma, Pinecone, Milvus, Weaviate
- **Web Technologies**: HTML, CSS, JavaScript, Streamlit
- **Cloud and DevOps**: Azure DevOps, Google Cloud, Amazon Web Services, Fast API, Docker, Kubernetes, Azure AI, Open AI, Amazon Bedrock
- **AI/ML**: Generative AI, NLP, LLMOps, MLOps, RAG, AutoGen, MCP, LangChain, LlamaIndex, LangGraph, Prompt Engineering, Computer Vison, Transformers, LoRa, Pandas, PySpark, PyTorch, TensorFlow, Keras

WORK EXPERIENCE

Calsoft Pvt Ltd | Data Science Intern | Pune, India — Apr 2024 - Jun 2025

- Executed the end-to-end project lifecycle within the core GenAI team, delivering scalable, AI-driven solutions from architecture to optimization.
- Built a multi-pipeline, agent-based RAG architecture, increasing test case relevance accuracy to **95%** and enabling retrieval of legacy test cases.
- Developed and deployed an on-premises AI solution using open-source LLMs, achieving an **80% reduction** in manual test case generation and significantly enhancing QA productivity.

Intellect Design Arena Ltd | AI/ML Intern | Pune, India — Sep 2023 - Mar 2024

- Engineered core components of IGTB Copilot, an internal project enabling Relationship Managers to leverage actionable insights from bank data.
- Implemented LLM integration and prompt engineering for a successful proof of concept Copilot, automating payment onboarding workflows by **over 70%**.

Intellect Design Arena Ltd | Software Developer Intern | Pune, India — Jan 2023 - Mar 2023

- Engineered Natural Language Processing techniques for efficient filtering of bank customer data.
- Developed a project utilizing Streamlit and AI tools, directly contributing to a **45% increase** in KYC check efficiency.

PROJECTS

Text Gen from Scratch | Codebase

- Engineered a **6-layer** transformer with custom positional encoding and learning rate scheduling to improve training convergence.
- Built an end-to-end ML pipeline for text generation with a **50k-word** vocabulary and **64-token** sequences, enabling scalable training across **50 epochs**.

Medi Match Copilot | Codebase

- Integrated Llama 3.1 for natural language processing and Llama 3.2-Vision for medical image analysis, supporting robust multimodal reasoning in clinical workflows and achieving **up to 95%** confidence in diagnostic assessments.
- Created a modular Python-based system for scalable retrieval of medical data, generation of diagnostic hypotheses, and visual analysis, tailored for long-term maintainability.
- Delivered a comprehensive clinical assistant automating patient symptom interpretation, medical image evaluation, and evidence-based knowledge synthesis, assisting in data-informed diagnostic decisions.

Eco Policy Copilot | Codebase

- Integrated advanced data modeling and workflow automation to support evidence-based decision-making for climate and economic policy initiatives, assisting environmental researchers and analysts.
- Constructed an AI-driven platform using LangChain and LangGraph, processing **6 data sources** and over **20 policy types**, analyzing climate, economic, and energy data, and delivering actionable policy recommendations for sustainable development.
- Designed an Agentic workflow, enabling autonomous agents to process data, simulate policies, and generate executive summaries for streamlined environmental analysis.

RESEARCH AND PUBLICATIONS

- "**Enhancing RAG performance through Chunking and Text Splitting Techniques"** published in the International Journal of Scientific Research in Computer Science, Engineering, and Information Technology. | Link